

LA5222 Field Journal — Desktop Study

Water at Bidadari Park: From National Catchment Policy to Alkaff Lake's Treatment Train

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Purpose of this note

This document assembles, moving from Singapore's national water-supply and drainage policy down to the specific engineering of Alkaff Lake and its terraced wetlands, what can be verified from official and academic sources about the abiotic water system that shapes the paired field patch at Bidadari Park. It is the water/abiotic counterpart to the companion desktop-study timeline on the site's history, planning, and socio-cultural layers, and is meant to be read alongside it. Five on-site interpretive panels (NParks and PUB signage at the park entrance and at Alkaff Lake), photographed and transcribed by the author on 5 September 2026, are used here as primary sources in their own right, in the same way the Alkaff Gardens Heritage Walk panel was used in the historical timeline. Where a detail could not be confirmed against a citable source – or could only be retrieved indirectly, for instance through a search-engine summary rather than a direct fetch of the originating PUB page – it is flagged as unconfirmed rather than stated as fact.

From national water policy to site-scale water design

Singapore scale: catchment policy and infrastructure

The Four National Taps Singapore's water supply rests on four sources developed as a deliberate diversification strategy: water from local catchment, imported water, high-grade reclaimed water (NEWater), and desalinated water (Public Utilities Board, 2026). The strategy traces back to the 1972 Water Master Plan, which envisioned this diversified portfolio over a roughly fifty-year horizon (Public Utilities Board, 2026; Irvine et al., 2014). A peer-reviewed account by Irvine, Chua and Eikass, co-authored with a PUB engineer, describes the Four National Taps as a holistic approach linking drainage engineering to drinking-water supply, from stormwater capture through to reclamation and desalination (Irvine et al., 2014). **Caveat:** the exact current-year percentage contribution of each tap changes as new plants and reservoirs come online; figures reported in secondary sources (for example, NEWater meeting roughly 40 percent of demand at present, with PUB targeting about 55 percent from NEWater and about 30 percent from desalination by 2060 against a demand projected to double) could not be verified against a directly fetched PUB page in this research session – attempts to fetch pub.gov.sg pages directly returned access errors, so the figures above should be treated as indicative and re-checked against PUB's current published numbers before being quoted as fact. *[Official, with flagged gap; Academic: Irvine, Chua and Eikass 2014]*

Local catchment and the reservoir network Rain falling on roughly two-thirds of Singapore's land area is collected through a network of drains, canals and rivers and channelled to the island's reservoirs, commonly reported as 17 in number, before treatment for potable supply (Public Utilities Board, 2026). This is stated in near-identical terms by the on-site PUB panel at Alkaff Lake itself: "Two-thirds of Singapore's land is water catchment. Rainwater that falls in these areas is collected through a network of drains, canals and rivers, and channelled to our 17 reservoirs" (Public Utilities Board, 2024c). PUB has stated an intention to expand catchment area further, toward 90 percent of land area by 2060, as demand grows (Public Utilities Board, 2026). **Caveat:** the reservoir count is a commonly repeated figure but may have changed since the sources consulted here were published; it is treated as the best-documented current figure rather than a fixed constant. *[Official: PUB; Primary/on-site: PUB interpretive panel, photographed on site]*

Marina Reservoir and Marina Barrage as flagship urban catchment Marina Barrage, completed and opened on 31 October 2008, dams the mouth of the Marina Channel to create Marina Reservoir, commissioned as a freshwater reservoir on 20 November 2010 (Public Utilities Board, 2024d). Its catchment spans about 10,000 hectares, described by PUB as the island's largest and most urbanised catchment, and its completion (together with the Punggol and Serangoon reservoirs) is what pushed Singapore's total water catchment from one-half to two-thirds of national land area (Public Utilities Board, 2024d). Beyond water supply, the barrage's nine crest gates and pumping station give it an explicit flood-control function, releasing stormwater to sea at low tide and pumping it out at high tide to protect low-lying downtown areas (Public Utilities Board, 2024d). This is directly relevant to Bidadari at site scale: the on-site catchment panel states plainly that "stormwater runoff flows from here [Alkaff Lake] into the Marina Reservoir," making Bidadari one small, documented contributing catchment within this larger, engineered urban water system (Public Utilities Board, 2024c). *[Official: PUB; Primary/on-site: PUB interpretive panel]*

The ABC Waters Programme PUB's Active, Beautiful, Clean Waters (ABC Waters) Programme was launched in 2006 to integrate drains, canals and reservoirs with the surrounding built environment, moving beyond a purely hydraulic conception of drainage infrastructure toward waterbodies that are simultaneously functional, aesthetic, and community-facing (Public Utilities Board, 2024b). A Centre for Liveable Cities policy study situates this within a broader set of institutional and regulatory changes through the 2000s that made such integration possible, framing Singapore's waterways as environmental assets rather than purely engineering assets (Centre for Liveable Cities, 2017). PUB's ABC Waters Design Guidelines, first issued in 2009 and updated in a 2024 edition, document a defined set of design features – constructed wetlands (surface-flow, floating, and subsurface types), vegetated swales, bioretention swales, bioretention basins (rain gardens), and sedimentation basins – intended to be combined in sequence so that runoff is progressively slowed, detained, and cleansed as it moves from where it falls toward the receiving waterbody (Public Utilities Board, 2024a). **Caveat:** this sequencing is referred to in this note, and on the Bidadari site panel itself, as a "treatment train"; the more general engineering framing

of pollutant movement as a “source-pathway-receptor” chain is standard vocabulary in the wider stormwater-management literature, but this note could not confirm that PUB’s own guidelines use that exact three-part terminology, so the source-pathway-receptor label should be read as an explanatory gloss rather than a direct PUB quotation. [Official: PUB; Academic/policy: Centre for Liveable Cities 2017]

Site scale: Alkaff Lake and the terraced wetlands

Alkaff Lake as a treatment-train retention pond The on-site PUB welcome panel states: “Alkaff Lake is situated within Bidadari Estate. Under PUB’s Active, Beautiful, Clean Waters (ABC Waters) programme, the lake now has enhanced greenery, plaza and viewing deck. This retention pond showcases a treatment train of ABC Waters design features of constructed wetlands and swales” (Public Utilities Board, 2024h). PUB’s own media factsheet for the lake’s opening confirms and refines the technical figures already flagged in the companion history document: the lake covers 1.8 hectares, holds up to 40,000 cubic metres of water at maximum (about 16 Olympic-sized pools, consistent with the figure cited from NParks and CPG Corporation) (National Parks Board, 2024b; CPG Corporation, 2024; Public Utilities Board, 2024e), but typically holds only around 10,000 cubic metres in dry weather, with a typical water level of 1–1.5 metres rising to a maximum of about 4 metres during storms (Public Utilities Board, 2024e). During heavy rain, runoff from roughly 43.5 hectares – about half of Bidadari estate – is diverted into the lake to regulate discharge into downstream drainage (Public Utilities Board, 2024e); in dry weather it functions as an ordinary recreational and aesthetic community space (Public Utilities Board, 2024e,h). Automatic water-level sensors with beacon lights and audio broadcasts are installed to warn the public before and during flood events (Public Utilities Board, 2024e). [Official: PUB; Primary/on-site: PUB interpretive panel]

The terraced wetlands: plant selection and function The first of two PUB “Terraced Wetlands” panels at the site lists the wetland plants selected specifically for their water-cleansing properties: Umbrella Papyrus (*Cyperus alternifolius*), Pickerelweed (*Pontederia cordata*), Narrowleaf Cattail (*Typha angustifolia*), and Alligator Flag (*Thalia geniculata*); the same panel notes that the damselfly Common Bluetail (*Ischnura senegalensis*) uses the wetland, and states that “the wetlands support a variety of plant and wildlife species. This promotes a healthy ecosystem, contributing to biodiversity and ecological balance that can be enjoyed by the public” (Public Utilities Board, 2024f). The second panel explains the underlying mechanism: “These play a crucial role in urban environments by capturing pollutants and managing water quality. As runoff comes into contact with the ground, it picks up sediments, nutrients and other impurities which would eventually flow into Alkaff Lake. Water from the lake is thus periodically recirculated through the wetlands, which help to filter out fine sediments and absorb pollutants” (Public Utilities Board, 2024g). Two “Did you know?” callouts on the same panel add that microorganisms growing on plant roots absorb nutrients and contaminants from runoff, and that the combination of physical filtration and

biological uptake in these constructed wetlands ensures cleaner water for Alkaff Lake, benefiting both environment and community (Public Utilities Board, 2024g). Read together, these two panels are effectively the site-level, plant-species-specific instantiation of the ABC Waters treatment-train concept described at national scale above. **Caveat:** the park's own map legend also lists "Ficus Hill Cascades" and "Ficus Hill Terrace" as distinct features (Public Utilities Board, 2024h); whether these correspond to the same cascading wetland system that CPG Corporation's project account calls the "Ficus Stream" (CPG Corporation, 2024) was not confirmed in the sources retrieved for this note and should be checked on site or with NParks/PUB before the two names are treated as synonymous. *[Primary/on-site: PUB interpretive panels; Official, with flagged naming gap: CPG Corporation]*

Recirculation and aeration Beyond the passive filtration described above, PUB's media factsheet notes that aerators are positioned within Alkaff Lake itself to improve water circulation and raise dissolved-oxygen levels (Public Utilities Board, 2024e), a mechanical complement to the plant- and microorganism-based filtration performed by the terraced wetlands and to the periodic lake-to-wetland recirculation described on the panel (Public Utilities Board, 2024g). *[Official: PUB; Primary/on-site: PUB interpretive panel]*

Gross Pollutant Trap (GPT) The park's on-site map legend lists a Gross Pollutant Trap alongside the Entrance Plaza, Viewing Deck, Rain Tree Island, Terraced Wetlands, Shelter, Ficus Hill Cascades, Ficus Hill Terrace, and bus stop as named features of the Alkaff Lake precinct (Public Utilities Board, 2024h), and PUB's media factsheet confirms that a Gross Pollutant Trap is installed at the site to trap litter and silt from incoming runoff (Public Utilities Board, 2024e). No Bidadari-specific technical description of the unit's capacity, make, or exact placement was located in the sources consulted. In general stormwater engineering terms, a GPT is a primary-treatment device, typically the first stage of a treatment train, designed to capture visible litter and coarse sediment (conventionally particles above about 5 millimetres) before they reach downstream wetlands or waterbodies; because GPTs are not effective at removing dissolved nutrients, they are normally paired with a secondary biological stage such as a constructed wetland, exactly the sequence documented on site at Bidadari (GPT upstream of the terraced wetlands and lake) (Public Utilities Board, 2024h,g). **Caveat:** this functional description is drawn from general urban-drainage engineering literature rather than a Bidadari-specific PUB or CPG technical document, and is flagged as such. *[Primary/on-site: PUB interpretive panel and map legend; general engineering context, not site-specific]*

ABC Waters status of the project PUB's own media factsheet states that Alkaff Lake "incorporates PUB's Active, Beautiful, Clean Waters (ABC Waters) design features such as terraced wetlands and swales," and the on-site welcome panel describes the lake as showcasing an ABC Waters "treatment train" (Public Utilities Board, 2024e,h). This confirms that the project was designed to, and does, adopt ABC Waters design principles. **Caveat:** no separate documentation of a formal "ABC Waters Certified" award or listing specifically for Alkaff Lake in PUB's

certified/showcase-project registry was located in this research session; the claim made here is limited to what PUB itself states (that the site incorporates ABC Waters design features), which is a narrower and better-supported statement than asserting formal certification. *[Official: PUB; Primary/on-site: PUB interpretive panel]*

Orientation and etiquette signage A general NParks orientation map at the park entrance, marked “You Are Here” near Alkaff Lake, sets out the park’s overall layout and visitor etiquette rules (National Parks Board, 2024a). While not itself a water-technical source, it is the panel that frames how a visitor (and this field researcher) locates the lake, the wetlands, and the other features described above within the wider park, and is recorded here for completeness and reproducibility of the desktop study’s primary-source base. *[Primary/on-site: NParks interpretive panel]*

Synthesis: connecting the water-technical material to the field journal

For the abiotic-data and planning-context strands of the desktop study, the material above shows that the field patch around Alkaff Lake is not an incidental water feature but a deliberately engineered node inside a national system: rain falling on the site is part of the two-thirds of Singapore’s land area treated as catchment, the lake’s stormwater function is explicitly wired into the same Marina Reservoir/Marina Barrage system that anchors national water security, and the lake-plus-wetlands complex is a documented, if not formally certified, instance of PUB’s ABC Waters treatment-train design philosophy (Public Utilities Board, 2026, 2024d,c,e). For the biotic strand, the two Terraced Wetlands panels give a ready-made, site-specific species list – Umbrella Papyrus, Pickerelweed, Narrowleaf Cattail, Alligator Flag, and the Common Bluetail damselfly – that can be cross-referenced directly against Daniela’s own photo-based plant classification taken elsewhere on site, turning the interpretive signage from background reading into a testable link between the abiotic water-treatment function documented here and the biotic community it is explicitly designed to support (Public Utilities Board, 2024f). Read together with the companion historical timeline, this suggests the field patch is best framed in the Field Journal not simply as a scenic lake, but as a layered site where engineered abiotic infrastructure (catchment, retention, filtration), planning process (a multi-agency PUB/NParks/HDB/NHB delivery), and biotic community (wetland flora selected for function, and the fauna it attracts) are all operating on the same few hectares at once.

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AI disclosure: This desktop-study note was compiled with the assistance of Claude (Anthropic), which transcribed primary-source content from the author's on-site photographs of NParks and PUB interpretive panels (5 September 2026) and conducted web-based research using official PUB sources, a Centre for Liveable Cities policy report, and a peer-reviewed journal article. All citations were checked against the retrieved source content where a direct fetch succeeded; where a PUB page could only be accessed indirectly through search-engine summarisation, or where a figure could not be verified at all, this is flagged explicitly in the text above rather than stated as fact. This draft should be checked against original sources before submission.